



Unit 4 · Outcome 3 · Scientific Investigation

Methodology & variables

KK02

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. 02 Methodology & variables VCAA study design · the examinable dot point

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



Methodology & variables

Unit 3 & 4 · Outcome 3 · Key Knowledge 02 Methodology & variables VCAA study design · the examinable dot point "Characteristics of the selected methodology and method, and the appropriate use of independent, dependent and controlled variables." 7 methodologies 3 live 3D models IV · DV · CV trinity 5 exam questions · self-mark

Methodology, method and technique are not the same word

The core distinction Methodology, method and technique are not the same word Examiners lose patience with students who use these three as synonyms, because they describe three different levels of decision . Nail the hierarchy and a whole family of questions becomes easy. LEVEL 1 · WHAT TYPE Methodology The overall type of scientific investigation you choose — e.g. a controlled experiment or fieldwork . VCAA lists seven for Environmental Science. LEVEL 2 · HOW Method The specific procedure that carries out that methodology — the ordered steps: what you do

One creek, seven ways to investigate it

Live model · methodology switcher One creek, seven ways to investigate it Priya's worry — "salinity is harming the plants at Mirrin Creek" — can be turned into any of the seven VCAA methodologies . Pick one below: the model rebuilds to show how that investigation would actually run, and the panel gives its defining characteristic, when to choose it, and how variables behave in it. 3D · WebGL Mirrin Creek — methodology switcher drag to orbit · pick a methodology ↓ Interactive 3D model unavailable in this view — every methodology's description still w

The holy trinity: independent, dependent, controlled

Live model · the variable isolator The holy trinity: independent, dependent, controlled A variable is any factor that can change or be changed in an investigation. In a controlled experiment there are three jobs a variable can have. Get the three straight and most of this dot point's marks are yours. INDEPENDENT · IV The one you change The single factor you deliberately vary across groups. The presumed cause . There should be exactly one. Priya's IV: salt concentration of the watering solution (0, 4, 8, 12, 16 g/L). DEPENDENT · DV The one you measure The

Operationalise: a variable isn't done until it has units

Where marks are won and lost Operationalise: a variable isn't done until it has units To operationalise a variable is to state exactly how it is changed or measured — with units, a range and (for the IV) the levels tested. Vague variables can't be marked because nobody could repeat the investigation from them. This single habit lifts answers from a 1 to full marks. Not operationalised IV: salt DV: plant growth CV: conditions How much salt? Growth of what, measured how, over how long? "Conditions" names nothing. Unrepeatable → unmarkable. Operationali

A control group is not a controlled variable

The most-confused pair in the course A control group is not a controlled variable They sound identical and mean opposite things. Mixing them up is one of the most common ways students drop marks on this dot point, so here it is side by side.

Controlled variable A factor kept constant across every group so it can't muddy the result.

There are usually several. Priya keeps light, temperature and water volume identical for all five salt groups.

Control group The baseline group that gets no treatment (or the "normal" level of the IV), used as the comparis

Only a controlled experiment lets you control the variables

The nuance examiners reward Only a controlled experiment lets you control the variables The IV–DV–CV language was built for controlled experiments, where you manipulate the IV. In the field you usually can't.

Knowing how variables behave in other methodologies is a higher-order mark — and a classic exam trap. Same variables, different power over them Controlled experiment You set the IV (you choose 0/4/8/12/16 g/L), measure the DV, and hold CVs constant . Cause-and-effect can be claimed. Correlational study You measure both variables as they already occur

Sort each statement into its job

◇ ACTIVE RECALL

Active recall · drag & drop Sort each statement into its job From Priya's investigation, drag each card into the correct bin. On a touch screen, tap a card then tap its bin. Hit Check to mark — green is right, red is wrong. Mirrin Creek experiment — which job does each play? 0 / 6 placed Methodology Independent var Dependent var Controlled var ✓ s ↺

Command-word decoder

◆ COMMAND WORDS

Read the question properly Command-word decoder The verb in the question sets how much you must write. Click a command word to see what it demands on a methodology-and-variables question, and a worked Mirrin Creek example.

Seven traps that quietly cost marks

⚠ EXAM TRAPS

Lose-the-mark moments Seven traps that quietly cost marks Each of these is a real pattern from examiner reports on investigation-design questions. open the fix.

P·E·L trainer — justify a methodology

P·E·L WRITING

Build the paragraph P·E·L trainer — justify a methodology Short-answer marks come from structure. P·E·L = Point (name and define) → Example (anchor in Mirrin Creek) → Link (tie back to the question). Type a justification below; the keyword meter lights up as you hit the ideas examiners look for. P oint E xample L ink Prompt: "Priya wants to test whether salinity causes reduced growth in native rushes. Justify why a controlled experiment is the appropriate methodology, and identify her independent and dependent variables." (4 marks)



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Practice questions

★ PRACTICE & SELF-MARK

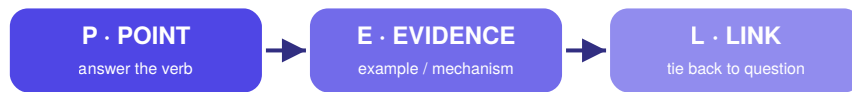
Exam-style · 20 marks Practice questions Five VCAA-style questions. For each, try it in your logbook first, then reveal the weak answer (and why it loses marks), the strong answer, and the marking guide . Mark yourself honestly — your score builds in the dock on the right. Self-marking honestly is the skill. The dock totals the marks you award yourself across all five questions (out of 20). Aim to justify every mark against the guide, the way an assessor would — generous marking only fools you in November.

Take it into the exam

One-screen recap Take it into the exam Methodology & variables — the essentials Methodology The type of investigation. The 7: classification & identification · controlled experiment · correlational study · fieldwork · modelling · product/process/system development · simulation. Method vs technique Method = the ordered procedure. Technique = the specific tool/skill (quadrat, EC meter, balance). Congruence rule The question chooses the methodology. Causal question → controlled experiment. Relationship in the field → correlational/fieldwork. IV The one fa

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Not operationalised IV: salt DV: plant growth CV: conditions How much salt? Growth of what, measured how, over how long? "Conditions" names nothing. Unrepeatable → unmarkable.

✓ STRONG / MODEL ANSWER

Operationalised IV: NaCl concentration of solution: 0, 4, 8, 12, 16 g/L DV: mean increase in shoot height (mm) over 21 days CV: light (12 h/day), 22 °C, 50 mL water/pot, same potting mix, same species & age Levels, units and a time-frame mean another scientist could reproduce it exactly. That's the bar.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.